

Lecture 34 — Embedded Real-Time Operating Systems

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This lecture is not yet written.

Intended sections:

- 1 What an RTOS is — and is not
- 2 Life without an MMU: flat address spaces and the MPU
- 3 Static allocation and why `malloc` is suspect
- 4 FreeRTOS anatomy: tasks, the tick, and the scheduler
- 5 Priority-based preemptive scheduling in practice
- 6 Synchronization: queues, semaphores, priority inheritance
- 7 Interrupts: deferring work out of the ISR
- 8 Tickless idle and low-power operation
- 9 Zephyr as the modern alternative
- 10 Choosing: embedded RTOS vs. real-time Linux